Integration of Quantum Key Distribution Networks and Classical Networks: An Evolution Perspective

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Abstract

As a promising quantum application, Quantum Key Distribution (QKD) technology has been rapidly developed from point-to-point communication to a large-scale commercial network. However, considering the cost of QKD components and network infrastructure, building a QKD network is challenged by its expensive capital expenditure, which significantly hinders its future popularization. In this article, we envision an evolution roadmap consisting of three stages for the future development of QKD networks, aiming at achieving a fully integrated network architecture that combines QKD and telecommunications networks. Powered by several critical technologies, such as link multiplexing, miniaturized QKD, and transmission optimization, QKD is foreseen to be integrated into telecommunications networks as an add-on module that can be flexibly deployed. After that, we introduce the integrated network architecture design in detail, and respectively provide four QKD-related function implementations. At last, key technologies that should be concerned and further investigated are also discussed with performance evaluation, which can provide theoretical guidance for both industries and academia. Based on the proposed roadmap and integrated architecture, we believe a scalable, easy-to-use, and affordable QKD service can be brought to current telecommunication networks, and extensively applied by ordinary people while using daily applications.

Introduction

Nowadays, Quantum Key Distribution (QKD), which can achieve information-theoretical secure key distribution, has been rapidly developed for decades and become one of the most mature quantum technology. To provide global quantum security for various Internet applications, large-scale QKD networks, e.g., SECOQC and integrated space-to-ground quantum communication network [1], are successfully constructed and progressively devoted to academia, financial business, and industry in the world.

Despite the fact that the secret-key rate of current QKD systems has been drastically increased from 500bps [2] to 110Mbps[1] [3], and the scale of QKD networks has been significantly expanded, a non-negligible but remaining challenge is expensive Capital Expenditure (CapEx) of QKD network infrastructure. For a typical QKD network, exclusive network infrastructure including QKD transceivers, private networking devices, and dedicated fiber links are indispensable. Considering the expensive components such as laser sources, detectors, and optical fiber, the cost per QKD unit is in the tens of thousands of dollars at least, and also the cost of fiber link is rapidly increasing as network expands [4]. Thus, from a cost-efficiency point of view, building a large-scale QKD network with exclusive infrastructure may not be a smart move.

Essentially, as an add-on technology [5] that provides information-theoretical secure keys, QKD is compatible with classical networks, such as widespread telecommunications networks. On the one hand, from the perspective of hardware infrastructure, quantum signals are first transmitted through *optical fibers* between two communication parties, then translated into classical random bits through *classical computing processors*, and finally distributed through interconnected *switches* in telecommunications networks during end-to-end key relaying process. Since these necessary equipments for QKD network, e.g., *optical fibers, classical computing processors,* and *switches*, are also required by telecommunications networks to connect arbitrary nodes and transmit classical data, it is a nature way to share the same fundamental infrastructure through multiplexing technology and etc. On the other hand, from the perspective of software protocol stack, no matter point-to-point QKD protocols such as decoy BB84 or end-to-end key relaying process, classical signaling based interactions are necessary. To achieve efficient and reliable data transmission, existing QKD systems generally invoke Internet protocols to achieve reliable transmission service, e.g., TCP/IP [5]. In this way, the functionality of classical signaling supported QKD communications can be easily extended to existing telecommunications

[1] The secret-key rate provided in this paper is recorded over 10km standard optical fiber.

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networks via building up an overlay network over Internet protocol suite.

Considering these compatible characteristics of QKD technology, one feasible development route for future QKD networks is to facilitate the integration of QKD networks and telecommunications networks. Some pioneering studies have already endeavored to merge QKD networks with optical networks through pervasive fiber infrastructure multiplexing and co-fiber transmission design [5], [6], [7], and thus quantum signals can be transmitted through shared fiber infrastructure in optical networks rather than dedicated fiber links in current QKD networks. However, these studies are preliminary attempts on the integration of physical layer and lack global vision from networking perspective. Thus, a long-term roadmap for future QKD development from both hardware and software perspectives remains blank.

To reach the ultimate goal of network integration between QKD networks and classical networks, in this article, we provide a promising solution from an evolution perspective. At first, starting from current QKD architecture as stated by standardization organizations, we envision an evolution roadmap for future network development instructions. Second, we design a fully integrated architecture based on Internet protocol suite for the integration of QKD networks and classical networks, and propose the specific implementation method. At last, based on the proposed architecture, we discuss key technologies possess direct influences on QKD and key relaying process, and further design trivial solutions with performance evaluation. Inspired by the proposed integrated architecture design from evolution perspective, we believe a scalable, easy-to-use, and affordable QKD service can be brought to telecommunications networks, and ubiquitous applications of QKD technology is foreseen to come true.

The main contributions of our work are summarized as follows.

- We propose a three-stage evolution roadmap from separated QKD networks to fully integrated QKD communication networks. The proposed roadmap provides step-by-step instructions for progressive network integration.
- We propose a novel integrated architecture with implementation details of QKD-related functions, and also analyze key technologies and challenges that should be addressed in practice.
- We conduct extensive simulations to evaluate the performance of different key distribution and traffic scheduling schemes, and results show the effectiveness of the proposed schemes and inspire future study for worthwhile optimization problems in the proposed architecture.

The rest of this article is organized as follows. In the following section we review current QKD network architecture and introduce an evolution roadmap. In the section "Integrated Architecture Design," we describe the integrated architecture design and the specific function implementation. After that, we discuss key technologies involving efficiency and security in detail, and finally conclusions are drawn in the section "Conclusion."

Evolution Roadmap for Integration Architecture

The Existing QKD Architecture: Standardization

The world's first conceptual network is DARPA QKD network deployed in October 2003 [2], from then on, short-range, metropolitan-coverage, and long-haul QKD networks are successively deployed with the increase of scalability and capability. As suggested by standardization organizations, e.g., ITU-T and IETF, the conceptual structure of QKD networks [5] can be illustrated in Fig. 1. This architecture consists of three logical layers as follows.

- *Infrastructure Layer:* This layer is constituted by physical QKD nodes and links, including quantum and public channels. Quantum channels handle qubit transmission, while public channels manage classical bit

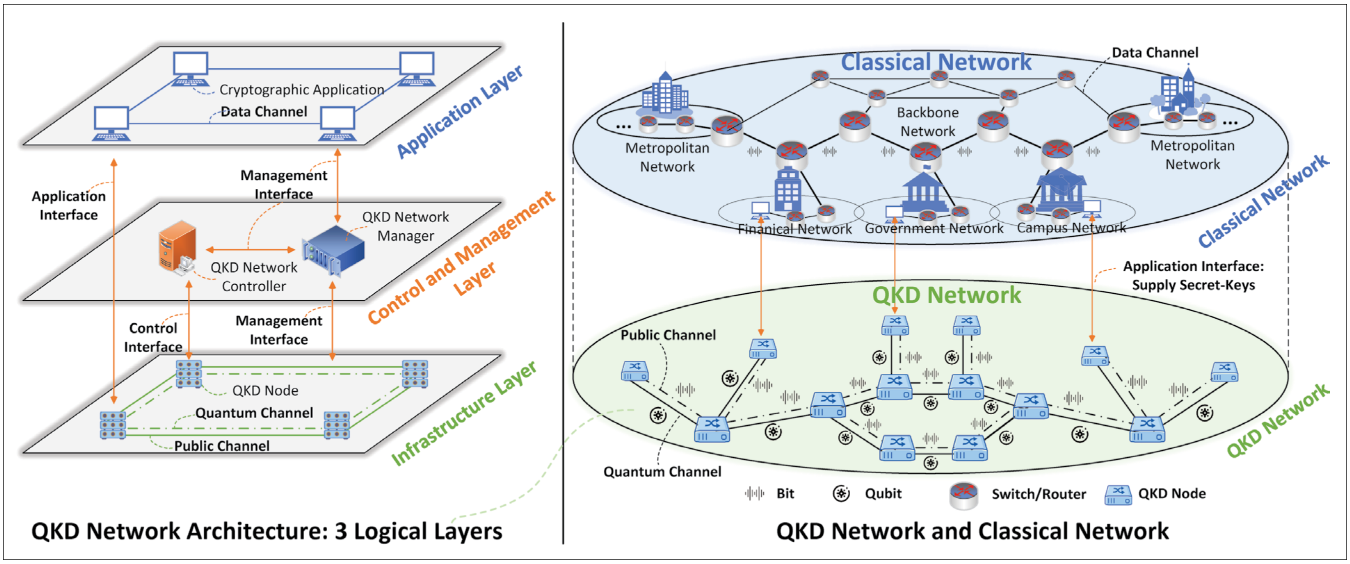

FIGURE 1. **The Existing QKD network Architecture**. Illustration of QKD networks and classical networks.

transmission. To achieve QKD process, a pair of QKD nodes apply prepare-and-measure approach[2] [5] through quantum channel, meanwhile, they also require necessary operations such as authentication, post-processing, and key relaying through classical channel.

- *Control and Management Layer:* This layer is constituted by QKD network controller and manager. All the QKD nodes are controlled by the controller, including routing control, multiplexing control, Quality of Service (QoS), etc. By contrast, the manager monitors and manages the QKD network as a whole, it monitors the status of QKD nodes and links, and supervises the controller. Its tasks may include fault, configuration, security management, etc [5], [8]. All interfaces, shown on the left side of Fig. 1, are established via classical channels.
- *Application Layer:* This layer is constituted by various cryptographic applications required by the users. According to specific requirements, the connected QKD node via application interface supplies secret-keys to the corresponding application, and then the application data can be encrypted using obtained secret-keys and transmitted over classical channel.

As shown on the right side of Fig. 1, QKD relies on telecommunications networks for interconnections to support authentication, base comparison, and post-processing. While current QKD network architectures have extended system reachability and availability, the high CapEx of dedicated QKD infrastructures remains a barrier to widespread adoption.

An Evolution Roadmap

As illustrated in Fig. 2, based on state-of-the-art QKD network architecture design and ubiquitous infrastructure of telecommunications networks, we envision an evolution roadmap for future development of QKD networks towards a fully integrated QKD communication network. This roadmap consists of three stages, which can be introduced as follows.

- *Stage 1: Separated QKD Networks and Telecommunications Networks*. At the time of writing, most QKD networks and telecommunications networks are existing independently, who respectively own physical infrastructure such as networking devices and fiber links. QKD-related signals, including quantum and classical signals, are transmitted through dedicated QKD networks, while classical application data is transmitted through commercial telecommunications networks, e.g., the Internet. These two networks are connected through exclusive application interface. For authorized users, they can send secret-key requests and implement encryption schemes once end-to-end key distribution is successfully completed. The key technology of this initial stage is to perfect QKD network architecture and increase the capacity of quantum communications. In currently deployed metropolitan area networks, the secret-key rate of QKD devices is typically 1~1000 Kbps [1, 8]. As a comparison, the capacity of networking devices in telecommunications networks, e.g., commercial Ethernet switches, is typically 1~10 Gbps. We are now located at this stage.
- *Stage 2: Coexisting QKD Communication Networks*. At this stage, QKD networks are foreseen to share the link-level hardware infrastructure of telecommunications networks including fiber optic cable (support qubit and bit transmission), coaxial cable (support bit transmission), optical line terminal, switches, etc. Both quantum and classical signals in QKD communications, i.e., single-photon and interaction signaling, and also data signals in cryptographic applications, are transmitted on the same link, e.g., optical fiber or free-space [5], as the one used in telecommunications networks[3], The symbolic event at this middle stage is widely

[2] Since entanglement-based approach is not mature as the time we writing, this article only considers prepare-and-measure approach.

[3] Most existing studies only consider co-fiber transmission for quantum signals and data signals. Here, we consider all signals, i.e., quantum signals via quantum channel, classical signals via public channel, and data signals via data channel are possibly transmitted over the same link.

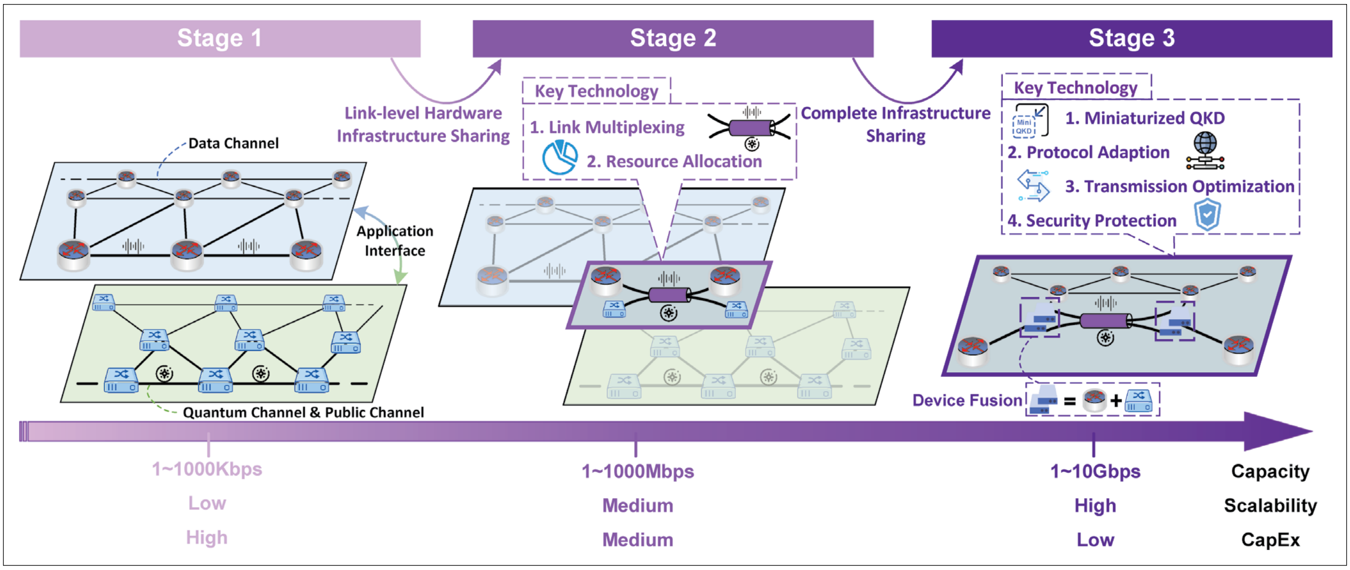

FIGURE 2. **An Evolution Roadmap**. Three-stage evolution roadmap from separated QKD networks to fully integrated QKD communication networks.

link-level hardware infrastructure sharing, and the key technologies are link multiplexing, e.g., Wavelength Division Multiplexing (WDM), and resource allocation in physical layer. The typical secret-key rate of QKD devices is predicted to reach 1~1000 Mbps.

- *Stage 3: Fully Integrated QKD Communication Networks*. At this stage, QKD networks are foreseen to be completely built upon the infrastructure of telecommunications networks. In addition to the link-level hardware infrastructure at stage 2, node-level hardware and software infrastructure are also leveraged by QKD networks through device fusion. Device fusion refers to customized QKD device, which is expected to be pluggable module such as transceiver [9], and corollary equipments such as optical switches, amplifier, and wave division multiplexer, can be installed into universal nodes including User Equipments (UEs) and universal networking devices. As a result, QKD devices can inherit existing protocols such as Internet protocol suite, and an overlay QKD network can be constructed over the physical network resource of telecommunications networks. Meanwhile, the control and management functions of QKD systems can be jointly optimized in telecommunications networks. The symbolic event at this final stage is widely complete infrastructure sharing, including node/link-level hardware and software, and the key technologies are miniaturized QKD, network protocol adaption, transmission optimization, and security protection in application layer. The typical secret-key rate of QKD devices is predicted to reach 1~10 Gbps.

Once the final stage is reached, QKD will thoroughly become an add-on and widely-deployed technology that can be easily enabled on existing networking devices. For example, users who require secure encryption method to protect real-time sessions, they can install separated QKD device on personal computers as needed rather than paying a large amount of money to connect exclusive but expensive QKD networks for secret-keys. Due to the significant decreasing of CapEx and scalability of such fully integrated QKD communication networks, we envision that QKD technology can benefit ordinary users for daily applications in the future, e.g., web browsing and video chatting. To introduce the construction method of fully integrated QKD communication networks, the specific architecture design is provided in the next section.

Integrated Architecture Design

Overview

Along with the development of hardware technology, e.g., multiplexing and miniaturized QKD, we envision that a fully integrated QKD communication network can be established via complete infrastructure sharing. The specific architecture design is depicted as Fig. 3. Based on the final stage of the proposed evolution roadmap, an overlay network[4] that supports QKD-related interactions between arbitrary nodes, e.g., post-processing signaling and key relaying process, is expected over the physical network resource of telecommunications networks. To provide all functions of current QKD networks, in this section, we introduce a specific architecture design that inherits widely deployed Internet protocol suite. By doing this, we believe a scalable, easy-to-use, and affordable QKD service can be brought to telecommunications networks and benefit ordinary people while using daily Internet applications.

QKD-Related Function Implementation

In existing QKD networks, as shown in Fig. 1, four basic functions are necessary, i.e., QKD process between adjacent nodes, key relaying process between remote nodes, network control and management, and authentication, each of which we introduce the specific implementation method in the proposed integrated architecture as follows.

1. QKD Process: For a typical QKD protocol, such as decoy BB84 [10], [11], 5 steps are necessary: 1. qubit preparation, transmission, and measurement, 2. sifting, 3. parameter estimation, 4. post-processing, 5. authentication. The first step requires qubit transmission via quantum channel, and the other four steps require local computation and bit transmission of classical signaling via public channel between adjacent nodes. Regardless of the QKD protocol used, QKD process is the basic function of QKD networks that implement QKD protocols and supply key materials between adjacent nodes. During the QKD process, the communication requirement can be classified as two categories, i.e., qubit transmission via quantum channel, and bit transmission of classical signaling via public channel. For the former one, the transceiver performs similar operations as the one in current QKD networks during qubit preparation and measurement process. For the latter one, interaction signaling can be transmitted by invoking mature TCP/IP interface, and TCP can provide reliable transmission service for adjacent nodes.

Due to hardware infrastructure sharing, multiplexing technology is generally required, we illustrate three possible scenarios for diverse signals transmitted on the same link in Fig. 4. In this way, joint transmission also introduces some potential challenges. When classical signals and data signals share the same link, joint transmission optimization must consider bandwidth competition and scheduling issues for these signals. On the other hand, when quantum signals and classical signals share the same link using WDM or similar multiplexing techniques, joint transmission optimization needs to account for physical layer impairments caused by co-fiber transmission, such as Raman scattering, four-wave mixing, and amplified spontaneous emission [12].

2. Key Relaying Process: Due to the limitation of physical-layer impairments, e.g., the scattering and loss of faint quantum signals transmitted in quantum channels, trusted relay is required to extend the communication distance in QKD networks. To achieve end-to-end key distribution between arbitrary remote nodes, key relaying is widely adopted. Regardless of the QKD protocol used, the basic idea of key relaying is that relay node combines independent secret-keys shared by adjacent QKD nodes through One-Time Pad (OTP) method, and then forwards the result to

[4] Actually, there are several methods to support similar functions in a fully integrated QKD communication network, such as dedicated protocol design and SDN-based realization supported by programmable switches. Limited by page length, we take overlay network as one promising solution.

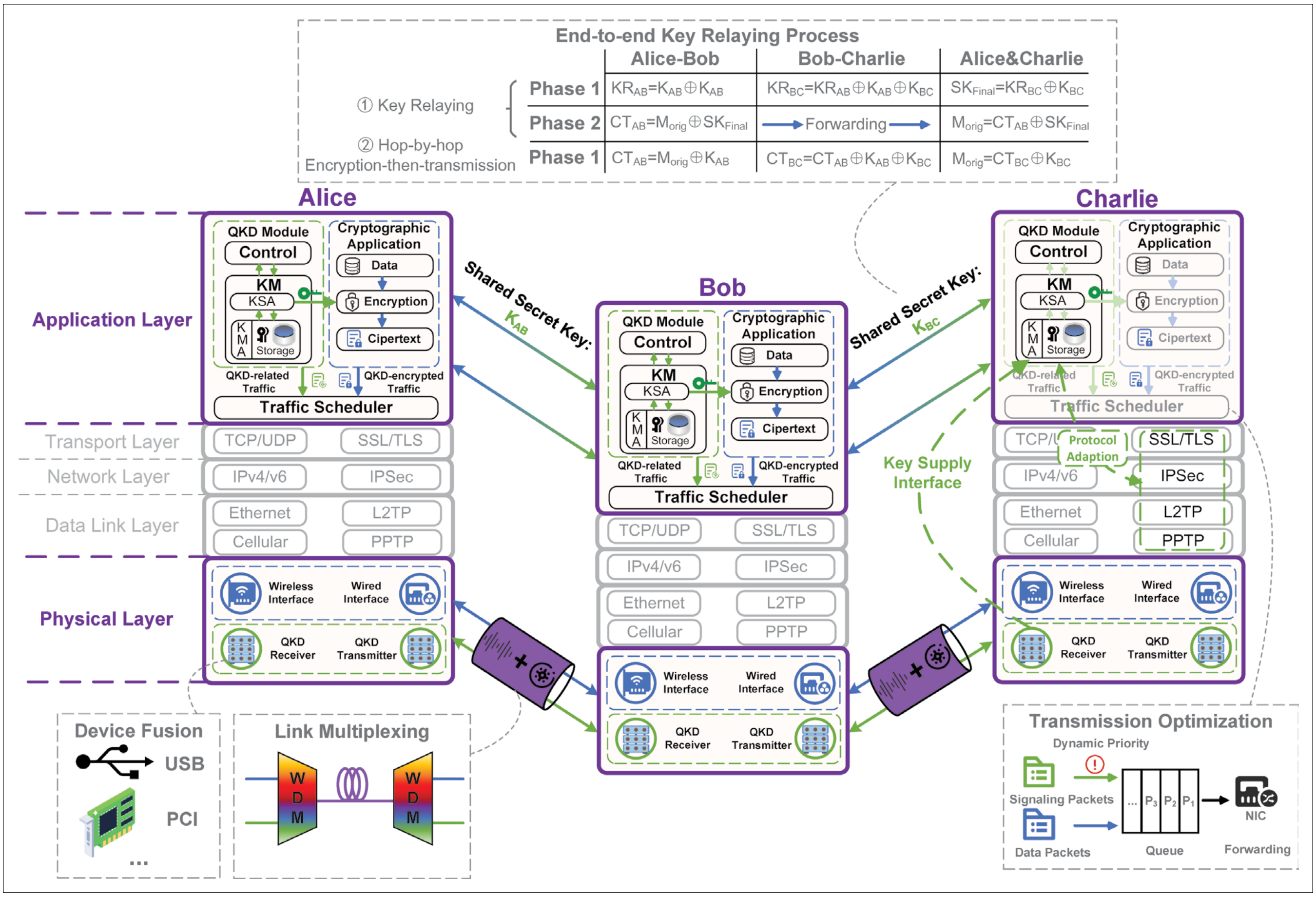

FIGURE 3. **Integration Network Architecture**. Illustration of the specific architecture design and corresponding key technologies.

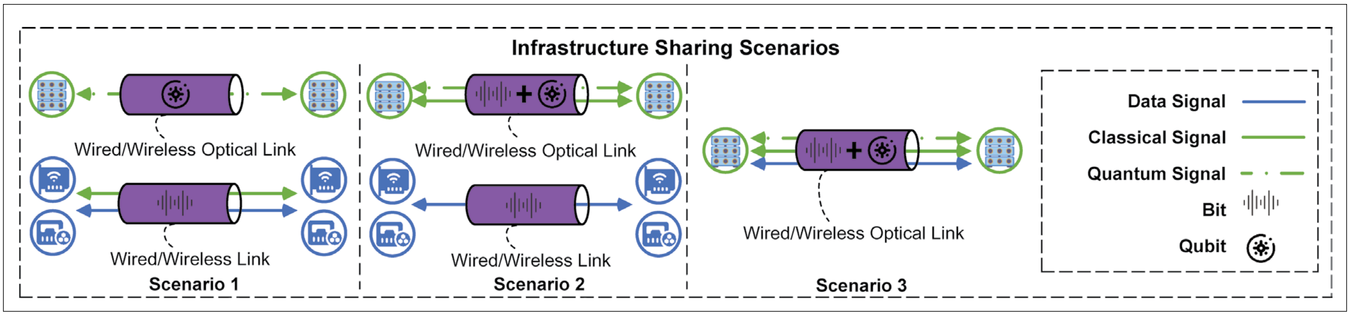

FIGURE 4. **Link-level Hardware Infrastructure Sharing**. Illustration of three different scenarios for coexisting signals on the same link.

the next node until reaching destination. As a result, the destination node can retrieve the original secret-key by performing a bitwise exclusive or operation (reflexive relation of exclusive or operation in OTP method) [5]. It is worth noting that in the process of key relaying, multipath transmission may occur. To ensure the correctness, the destination node must reorganize the keys that arrive in sequence before reconstructing the original key.

In essence, the process of key relaying constitutes classical computing (i.e., bitwise exclusive or operation) and classical transmission. As shown in Fig. 3 (see upper right diagram), secret-keys K_{AB} and K_{BC} are respectively produced via Alice-Bob link and Bob-Charlie link, and stored in secure key storage on UEs. After bitwise exclusive or operation $K_{AB} \oplus K_{BC}$ provided by encryption module on Bob, the encrypted data can be transmitted through classical channel to Charlie, and the original secret-key K_{AB} can be obtained via $K_{BC} \oplus (K_{AB} \oplus K_{BC})$ by Charlie.

Alternatively, in the proposed integrated architecture, a simple but effective method named hop-by-hop encryption-then-transmission is preferred rather than key relaying in current QKD networks. The basic idea is that key materials shared by adjacent nodes are directly consumed during data transmission on the overlay network without extra key relaying process, each intermediate node repeats operations of decryption and encryption-then-transmission until destination node. Even during multipath transmission, there is no need for additional reorganization of keys. Instead, multipath transmission methods from traditional networks can be directly applied in hop-by-hop encryption-then-transmission method. The goal of key relaying is sharing secret-keys between two remote nodes and the cryptographic application of both can utilize the shared keys to achieve end-to-end data encryption/decryption. However, in the proposed integrated architecture, hop-by-hop encryption-then-transmission method can be easily applied, and improve the transmission efficiency while facing packet loss in telecommunications networks. One example is shown in Fig. 3 (see upper right diagram), Alice encrypts the original data M_{orig} locally using the OTP method and sends the ciphertext CT_{AB} to Bob. After data decryption using shared key K_{AB}, Bob repeat local data encryption by using shared key K_{BC} and transmit the cipertext CT_{BC} to Charlie.

The performance comparison of key relaying and hop-by-hop encryption-then-transmission is conducted in Fig. 5 through a discrete-event driven quantum network simulator, called SimQN [13]. In the simulation, a dumbbell topology is adopted, and an amount of application metadata (1Gbits) is consistently sending out from source to destination (the distance is 3 hops) with encryption requirement[5]. Since key relaying method consumes double capacity to achieve end-to-end key distribution and encrypted data transmission, results show that hop-by-hop encryption-then-transmission in the integrated architecture has significant superiority with 50% performance improvement in terms of average throughput and Flow Completion Time (FCT).

3. Network Control and Management: In the integrated QKD architecture, as shown in Fig. 3, the add-on QKD module in application layer establishes an overlay network over the telecommunications networks, which is similar to peer-to-peer overlay networks. The network control and management can be realized in a centralized or decentralized manner depending on application scenarios. For example, in a private network scenario, Software-Defined Networking (SDN) is preferred to achieve flexible control and management through flow table update. In opposite, decentralized approach is a better choice due to the advantages of signaling delay and robust in a large-scale network scenario.

4. Authentication: Authentication is an important aspect in the process of communications between two communication parties since it guarantees the identity legitimacy, especially in the process of quantum key distribution. All existing authentication solutions can be applied in the proposed integrated network and we introduce as three of them as follow. Conventional encryption and authentication methods, such as Diffie-Hellman key exchange and digital signatures, have been extensively applied in telecommunications networks. However, these methods rely on difficulties in mathematical problem solving, and thus cannot provide provable security and become vulnerable against future quantum computing attacks. Another solution is pre-shared key authentication, which is mainstream authentication method in the existing QKD systems [5]. The basic idea is to pre-share a small amount of symmetric seed keys, and then respectively encrypt and decrypt the hash value of classical data. This method indeed can guarantee the

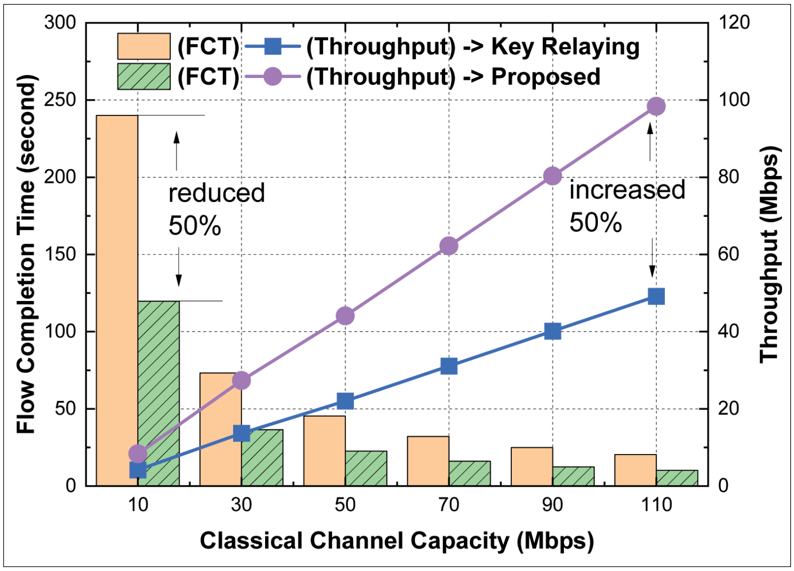

FIGURE 5. Performance comparison between two different end-to-end key distribution methods in the integrated architecture (Key Relaying vs. Hop-by-hop Encryption-then-transmission).

information theoretical secure authentication, but the burden of practical pre-shared operations becomes heavy and inefficient when the scale of network expands. Moreover, one prospective method, Post-Quantum Cryptography (PQC) based authentication, is also considered in recent years [14]. In essence, it is an improved conventional authentication method that relies on novel mathematical problems. While it serves as a short-term solution, and its long-term security remains questionable.

To be noticed, all authentication methods require interactions via shared communication channels. According to the specific security requirement of various applications, the aforementioned authentication methods can be carefully investigated and applied.

Open Issues

Although the proposed integrated architecture can fully leverage the infrastructure and significantly mitigate the cost of current QKD networks, still, there are some open issues should be further addressed, e.g., joint transmission optimization and security protection. The former one considers the efficiency problem in the proposed network. The latter one considers the security problems that arise from heterogeneous devices.

1. Joint Transmission Optimization: To generate raw key materials between two adjacent nodes, not only qubit transmission is necessary, but also interactions for the verification of exchanged key material are required in prepare-and-measure approach, i.e., interactions in post-processing, authentication steps. However, how to achieve efficient transmission coexisting with QKD-related traffic and application data traffic becomes an critical but open issue. To exemplify this transmission issue, as shown in Fig. 3, a traffic scheduler is deployed at the bottom in application layer. Except other non-cryptographic applications traffic, two kinds of traffic, i.e., QKD-related traffic and application

[5] The experimental setup and code can be accessed from https://github.com/QNLab-USTC/QKDN-hop-by-hop-encryption-then-transmission.

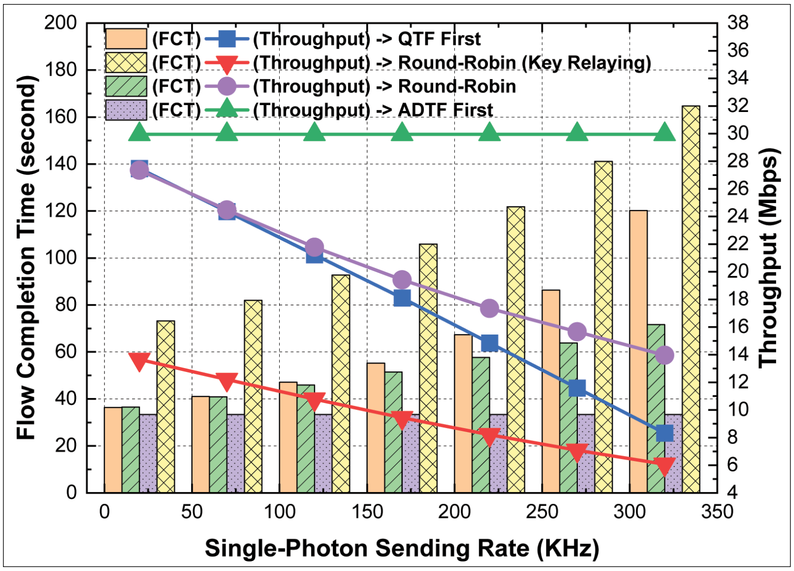

FIGURE 6. Performance comparison by considering joint transmission optimization (Prioritized Scheduling versus Fair Scheduling).

data traffic, are coexisting in the network and go through the scheduler. Since the generation process of symmetric key material requires both qubit and bit transmission, which involves the verification process of the exchanged key material, thus, the performance of QKD-related traffic transmission determines the performance of secret-key rate. Once QKD-related traffic is delayed, then the secret-key rate of raw key materials can be degraded. Moreover, since the QKD-related traffic is hundreds of times greater than the secret-key rate, it occupies a significant portion of the bandwidth. This, in turn, causes delays in encrypted application traffic within the integrated architecture, impacting the overall performance of QKD networks.

To address this issue, a joint transmission optimization is required. Here, we evaluate the performance of four trivial scheduling schemes and take hop-by-hop encryption-then-transmission as default, i.e., round-robin, QKD-related Traffic First (QTF), QKD-encrypted Application Data Traffic First (ADTF), and also adopt round-robin for key relaying method as a baseline. These schemes are deployed in the traffic scheduler as shown in Fig. 3, which can dynamically adjust the priority of traffic transmission via the identification of various traffic. We also conduct simulations in SimQN [13], and results in Fig. 6 show that QKD-encrypted application traffic first scheme can reach the highest throughput and FCT. In other schemes, as the sending rate of single-photon emitters increases, the QKD-related traffic rises sharply. Under limited link bandwidth, these results in delays in application data traffic, subsequently affecting throughput and FCT. In practice, the situation can be much more complicated in practical implementation, the fairness of various cryptographic applications, the key provisioning of burst traffic transmission, and real-time awareness of key storage on different nodes along the routing path should be jointly considered in the specific scheduler design.

2. Security Protection: The most attractive advantage of QKD technology is unconditional security of key exchange process. However, once the pluggable QKD transceiver module is installed on the universal device, after the key exchange process through prepare-and-measure approach, raw key material is stored on the local memory and can be easily accessed by attackers through malicious applications.

To diminish the potential security threat, the key storage and access method of raw key material should be further protected. At first, the dedicated QKD application can exchange symmetric key through pre-shared key between adjacent nodes, all raw key material can be stored after extra encryption through symmetric encryption algorithm, e.g., AES. To prevent the exposure of the original secret-keys during the key relaying process, the source node can encrypt them using pre-shared keys obtained from trusted Public Key Infrastructure (PKI) or PQC algorithms [14]. Additionally, secret-sharing methods can be employed, whereby the original secret-key is divided into multiple subkeys and transmitted via multiple paths [15]. As an attacker must obtain a sufficient number of subkeys to reconstruct the original key, this method ensures security even in the face of attacks on individual relay nodes.

3. Others: There are other open issues listed in the section "An Evolution Roadmap," such as link-level hardware infrastructure sharing, protocol adaption, and miniaturized QKD. Specifically, link-level hardware infrastructure sharing involves link multiplexing techniques such as WDM for scenario 2 and 3 as shown in Fig. 4, and critical challenges such as Raman scattering noise, four-wave mixing, and amplified spontaneous emission should be considered. Meanwhile, according to the variation of bandwidth requirements from various applications, dynamic resource allocation of physical communication resource is also necessary. By deploying efficient algorithm and providing hardware support, the bandwidth requirement can be guaranteed and the utilization efficiency of link-level hardware infrastructure can be improved. Protocol adaption refers to the application of key material into existing network protocols such as SSL/TLS, IPSec [5] as shown in Fig. 3, and necessary modifications to existing protocol design is required. Along with the future development of QKD networks following the proposed evolution roadmap, these issues should be further investigated from the perspective of both theoretical analysis and engineering application.

Conclusion

In this article, we first reviewed the existing QKD architecture, and analyzed the most challenging problem that hinders the future development of QKD networks, i.e., expensive capital expenditure of exclusive infrastructure. To address this practical issue, based on rapidly developing technologies such as link multiplexing and miniaturized QKD, we envisioned an evolution roadmap for future development of QKD networks towards an integrated QKD communication network. After that, we proposed the specific integrated network architecture design, and introduced the implementation method of four major functions existing in current QKD networks. Moreover, considering the potential issues leading by integration, we further discussed key technologies that should be further investigated in future integrated architecture, e.g., QKD/classical joint transmission optimization and security protection, and designed trivial solutions with performance evaluation. In a nutshell, we believe the proposed evolution roadmap and integrated network architecture can provide a promising proposal for future QKD network development.

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